

Parameter Effects on CV Prep Fidelity and Final Fidelity

1) What is being optimized

With `use_fock_expansion=True`, the CV input is treated as an incoherent mixture:

$$\rho_{\text{in}} \approx \sum_n |C_n|^2 |n\rangle\langle n|.$$

So the post-selected DV output is generally mixed:

$$\rho_{\text{post}} = \frac{1}{p_{\text{post}}} \text{Tr}\{\rho_{\text{CV}} \text{Left}[(\phi \rangle \langle \phi | \otimes I), U, (\rho_{\text{in}} \otimes |b\rangle\langle b|), U^{\dagger} \text{right}].$$

Use these metrics:

- Final fidelity (mixed-state correct): $(F = \langle \hat{u} | \rho_{\text{post}} | \hat{u} \rangle, (\hat{u} = u_{\text{theory}} / |u_{\text{theory}}|))$.
- PDE-vector match (amplitude-sensitive): $(|u_{\text{theory}}| - \sqrt{p_{\text{post}}}, \psi_{\text{principal}} / |u_{\text{theory}}|)$.

2) How parameters enter theoretically

LCHS coefficient shape (from your quadrature implementation):

$$C_n \sim \int dk; e^{-\gamma k^2}, g_{\beta_k}(k), \frac{H_n(k/e^{r'})}{\sqrt{2^n n!}}, \quad \gamma = e^{-2r'} - e^{-2r},; r' < r.$$

- `r_target = r`: sets post-selection squeeze and enters γ .
- `r_prime = r'`: sets prep-basis squeeze and γ ; also controls Hermite argument scaling.
- `kernel_beta`: shapes kernel factor $(g_{\beta_k}(k) = \exp((1+ik)^{\beta_k} / (1-ik)))$.
- `alpha_disp`: global scale of effective DV generator in evolution block.
- `energy_shift`: shifts identity term in generator (in this post-selected CV-DV map it changes overlap/success, not just a harmless phase).

- **beta** (in current run mode): mainly affects Gaussian-reference comparison ($|\langle \psi_{\text{lchs}} | \psi_{\text{gauss}} \rangle|^2$), not the fock-mixture evolution itself.

3) Observed trends from your sweeps (coarse grid)

From **sensitivity_results/*.csv**:

- **kernel_beta**: increasing **0.4** $\rightarrow$ **1.2** dropped final fidelity $\sim 0.64 \rightarrow \sim 0.26$.
- **r_prime**: increasing **0.3** $\rightarrow$ **1.5** dropped final fidelity $\sim 0.55 \rightarrow \sim 0.21$ and reduced post-probability strongly.
- **r_target**: weak/non-monotone; broad plateau in coarse scan.
- **alpha_disp**: increasing **0.6** $\rightarrow$ **1.4** improved final fidelity $\sim 0.34 \rightarrow \sim 0.53$.
- **energy_shift**: more negative improved fidelity (**-1.0** best in coarse scan).
- Coupled (**r_target, r_prime**): best region is low **r_prime** with moderate **r_target**, respecting $r' < r$.

Your latest tuned run achieved:

- Final mixed-state fidelity: **0.9509**
- Purity: **0.9489** (still mixed)

Interpretation: direction match is high, but amplitude-level vector mismatch is still nontrivial.

4) What can still be improved (feasible in this project)

1. Optimize a multi-objective score, not fidelity alone: $[J = F - \lambda \frac{\|u_{\text{theory}} - \sqrt{p_{\text{post}}} \psi_{\text{principal}}\|}{\|u_{\text{theory}}\|} - \mu, \max(0, p_{\text{min}} - p_{\text{post}})]$.
2. Do local joint scans around current optimum (**r_target, r_prime, alpha_disp, energy_shift**) instead of 1D-only sweeps.
3. Keep **r_prime** low and strictly below **r_target**; this is the most consistent high-fidelity region in your data.

4. Keep checking purity and post-selection probability with fidelity; mixed outputs can hide vector errors.